

Output Screenshots:

```
--- Holiday Tour Package Application ---
1. Add Holiday Package Details
2. Display All Holiday Package Details
3. Search Holiday Package By Id
4. Calculate Holiday Package Cost by Id
5. Exit
Enter your choice: 1
Enter Holiday Package Id: PK12345
Enter Source Place: Coimbatore
Enter Destination Place: Goa
Enter Number of Days: 4
Enter Basic Fare: 3000
Holiday Package added successfully.
```

```
--- Holiday Tour Package Application ---
1. Add Holiday Package Details
2. Display All Holiday Package Details
3. Search Holiday Package By Id
4. Calculate Holiday Package Cost by Id
5. Exit
Enter your choice: 2
PackageId: PK12345
Source: Coimbatore
Destination: Goa
No. of Days: 4
Basic Fare: 3000.0
Package Cost: 0.0
```

```
--- Holiday Tour Package Application ---
1. Add Holiday Package Details
2. Display All Holiday Package Details
3. Search Holiday Package By Id
4. Calculate Holiday Package Cost by Id
5. Exit
Enter your choice: 4
Enter Package Id: PK12345
Package Cost for PK12345 is: Rs. 13440.00
```

```
--- Holiday Tour Package Application ---
1. Add Holiday Package Details
2. Display All Holiday Package Details
3. Search Holiday Package By Id
4. Calculate Holiday Package Cost by Id
5. Exit
Enter your choice: 2
PackageId: PK12345
Source: Coimbatore
Destination: Goa
No. of Days: 4
Basic Fare: 3000.0
Package Cost: 13440.0
```

```
--- Holiday Tour Package Application ---
1. Add Holiday Package Details
2. Display All Holiday Package Details
3. Search Holiday Package By Id
4. Calculate Holiday Package Cost by Id
5. Exit
Enter your choice: 5
Thank You !!!
```

### CustomException

```
package com.holiday.CustomException;
public class InvalidHolidayIdException extends Exception {
    public InvalidHolidayIdException(String msg) {
        super(msg);
    }
}
```

### DaoInterface:

```
package com.holiday.Dao;
import java.util.List;
import com.holiday.model.*;
public interface HolidayDao {
    void addPackage(HolidayModel pkg);
    List<HolidayModel> getAllPackages();
    HolidayModel getPackageById(String packageId);
    void updatePackage(HolidayModel pkg);
}
```

DaoImplementation:

```
package com.holiday.Dao;
import java.util.ArrayList;
import java.util.List;
import com.holiday.model.*;
public class HolidayDaoImp implements HolidayDao {
    private List<HolidayModel> packageList = new ArrayList<>();
    @Override
    public void addPackage(HolidayModel pkg) {
        packageList.add(pkg);
    }
    @Override
    public List<HolidayModel> getAllPackages() {
        return packageList;
    }
    @Override
    public HolidayModel getPackageById(String packageId) {
        for (HolidayModel pkg : packageList) {
            if (pkg.getPackageId().equals(packageId)) {
                return pkg;
            }
        }
        return null;
    }
    @Override
    public void updatePackage(HolidayModel pkg) {
        for (int i = 0; i < packageList.size(); i++) {
            if (packageList.get(i).getPackageId().equals(pkg.getPackageId())) {
                packageList.set(i, pkg);
                break;
            }
        }
    }
}
```

Model:

```
package com.holiday.model;
public class HolidayModel {
    private String packageld;
    private String sourcePlace;
    private String destinationPlace;
    private int noOfDays;
    private double basicFare;
    private double packageCost;
    public HolidayModel(String packageld, String sourcePlace, String destinationPlace, int noOfDays, double
    basicFare) {
        this.packageld = packageld;
        this.sourcePlace = sourcePlace;
        this.destinationPlace = destinationPlace;
        this.noOfDays = noOfDays;
        this.basicFare = basicFare;
    }
    public String getPackageld() { return packageld; }
    public void setPackageld(String packageld) { this.packageld = packageld; }
    public String getSourcePlace() { return sourcePlace; }
    public void setSourcePlace(String sourcePlace) { this.sourcePlace = sourcePlace; }
    public String getDestinationPlace() { return destinationPlace; }
    public void setDestinationPlace(String destinationPlace) { this.destinationPlace = destinationPlace; }
    public int getNoOfDays() { return noOfDays; }
    public void setNoOfDays(int noOfDays) { this.noOfDays = noOfDays; }
    public double getBasicFare() { return basicFare; }
    public void setBasicFare(double basicFare) { this.basicFare = basicFare; }
    public double getPackageCost() { return packageCost; }
    public void setPackageCost(double packageCost) { this.packageCost = packageCost; }
    @Override
    public String toString() {
        return "Packageld: " + packageld +
            "\n Source: " + sourcePlace +
            "\n Destination: " + destinationPlace +
            "\n No. of Days: " + noOfDays +
            "\n Basic Fare: " + basicFare +
            "\n Package Cost: " + packageCost;
    }
}
```

#### ServiceInterface:

```
package com.holiday.service;
import com.holiday.model.*;
import com.holiday.CustomException.*;
import java.util.List;
public interface HolidayService {
void addPackage(HolidayModel pkg) throws InvalidHolidayIdException;
List<HolidayModel> fetchAllPackages();
HolidayModel findPackageById(String packageId);
double calculatePackageCost(String packageId) throws InvalidHolidayIdException;
}
```

#### ServiceImplementation:

```
package com.holiday.service;
import com.holiday.Dao.*;
import com.holiday.model.*;
import com.holiday.CustomException.*;
import java.util.List;
public class HolidayServiceImp implements HolidayService {
    private HolidayDao dao;
    public HolidayServiceImp() {
        dao = new HolidayDaoImp();
    }
    private boolean isValidPackageId(String packageId) {
        return packageId != null && packageId.length() == 7;
    }
    @Override
    public void addPackage(HolidayModel pkg) throws InvalidHolidayIdException {
        if (!isValidPackageId(pkg.getPackageId()))
            throw new InvalidHolidayIdException("Invalid Holiday Package Id");
        dao.addPackage(pkg);
    }
    @Override
    public List<HolidayModel> fetchAllPackages() {
        return dao.getAllPackages();
    }
    @Override
    public HolidayModel findPackageById(String packageId) {
        return dao.getPackageById(packageId);
    }
    @Override
    public double calculatePackageCost(String packageId) throws InvalidHolidayIdException {
        if (!isValidPackageId(packageId))
            throw new InvalidHolidayIdException("Invalid Holiday Package Id");
        HolidayModel pkg = dao.getPackageById(packageId);
        if (pkg == null)
            throw new InvalidHolidayIdException("Invalid Holiday Package Id");
        return pkg.getCost();
    }
}
```

```
        throw new InvalidHolidayIdException("Holiday Package Id does not exist");
        double basicFare = pkg.getBasicFare();
        int noOfDays = pkg.getNoOfDays();
        double cost = basicFare * noOfDays;
        double discount = 0.0;
        if (noOfDays > 10)
            discount = cost * 0.07;
        else if (noOfDays > 8)
            discount = cost * 0.05;
        else if (noOfDays > 5)
            discount = cost * 0.03;
        double amountAfterDiscount = cost - discount;
        double gst = amountAfterDiscount * 0.12;
        double finalCost = amountAfterDiscount + gst;
        pkg.setPackageCost(finalCost);
        dao.updatePackage(pkg);
        return finalCost;
    }
}
```

Main:

```
package com.holiday.main;
import com.holiday.service.*;
import com.holiday.CustomException.*;
import com.holiday.model.*;
import java.util.*;

public class HolidayMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HolidayService service = new HolidayServiceImp();
        while (true) {
            System.out.println("\n--- Holiday Tour Package Application ---");
            System.out.println("1. Add Holiday Package Details");
            System.out.println("2. Display All Holiday Package Details");
            System.out.println("3. Search Holiday Package By Id");
            System.out.println("4. Calculate Holiday Package Cost by Id");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            int choice = sc.nextInt(); sc.nextLine();
            switch (choice) {
                case 1:
                    try {
                        System.out.print("Enter Holiday Package Id: ");
                        String id = sc.nextLine();
                        System.out.print("Enter Source Place: ");
                        String src = sc.nextLine();
                        System.out.print("Enter Destination Place: ");
```

```
String dest = sc.nextLine();
System.out.print("Enter Number of Days: ");
int days = sc.nextInt();
System.out.print("Enter Basic Fare: ");
double fare = sc.nextDouble();
HolidayModel pkg = new HolidayModel(id, src, dest, days, fare);
service.addPackage(pkg);
System.out.println(" Holiday Package added successfully.");
} catch (InvalidHolidayIdException e) {
    System.out.println(e.getMessage());
}
break;
case 2:
    List<HolidayModel> list = service.fetchAllPackages();
    if (list.isEmpty())
        System.out.println("No Holiday packages available.");
    else {
        for (HolidayModel p : list) {
            System.out.println(p);
        }
    }
    break;
case 3:
    System.out.print("Enter Package Id: ");
    String sid = sc.nextLine();
    HolidayModel res = service.findPackageById(sid);
    if (res == null)
        System.out.println(" Holiday Package not found.");
    else System.out.println(res);
    break;
case 4:
    System.out.print("Enter Package Id: ");
    String pid = sc.nextLine();
    try {
        double cost = service.calculatePackageCost(pid);
        System.out.printf("Package Cost for %s is: Rs. %.2f\n", pid, cost);
    } catch (InvalidHolidayIdException e) {
        System.out.println(e.getMessage());
    }
    break;
case 5:
    System.out.printf("Thank You !!!");
    sc.close();
    System.exit(0);
default:
    System.out.println("Invalid choice. Try again.");
}
}
}}
```